526_532_PHY_SCI_NA_S_CH21_L1_668304.indd Page 529 7/5/19 2:36 AM f-0256/152/GO02887/Inspire_Science_Physical/NA/SE/Inspire_Science_Physical/G1/007_66830 ...

How Substances Dissolve

Fruity drinks made from powdered mixes are examples of solutions made by dissolving solids in liquids. Like the hummingbird food shown in **Figure 1,** they contain sugar, as well as other substances, such as food coloring and added flavors.

The dissolving of a solid in a liquid occurs at the surface of the solid. Interactions of matter at the bulk scale are determined by electrical forces within and between atoms. To understand how water solutions form, keep in mind two things that you have learned about water. Like the particles of any substance, water molecules are constantly moving, and water is a polar molecule.

Get It?

Identify where a solid actually dissolves when placed in a liquid.

How it happens

Figure 4 shows the process of sugar dissolving in water. In step 1, the negative ends of water molecules are attracted to the positive ends of sugar crystals as water moves past the surface of the solid sugar. In step 2, the water molecules pull the sugar molecules into solution. Finally, in step 3, the water molecules and the sugar molecules mix evenly.

The process described in the three steps shown in **Figure 4** repeats as layer after layer of sugar molecules moves away from the crystal. The same three steps occur for any solid solute dissolving in a liquid solvent.

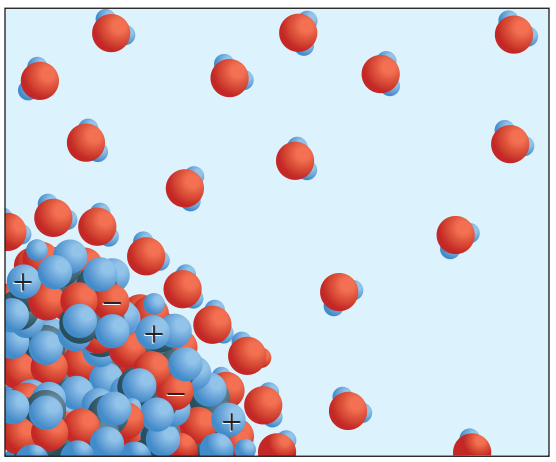
Dissolving liquids and gases

A similar but more complex process takes place when liquids and gases dissolve. Liquid and gas particles move much more freely than particles of solids move. When gases dissolve in gases or when liquids and gases dissolve in liquids, particle movement eventually spreads solutes evenly throughout the solvent, resulting in a homogeneous mixture.

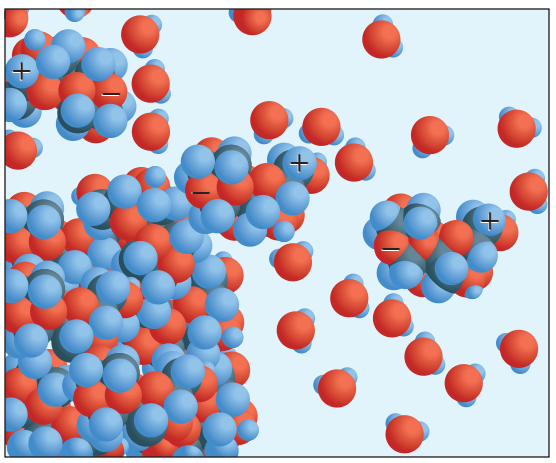
Dissolving solids in solids

Solid particles do move a little, but this motion is not enough to spread particles evenly throughout a mixture. Solid metals are first melted and then mixed together while still molten. In the liquid state, the atoms can spread out evenly and will remain mixed after they have cooled.

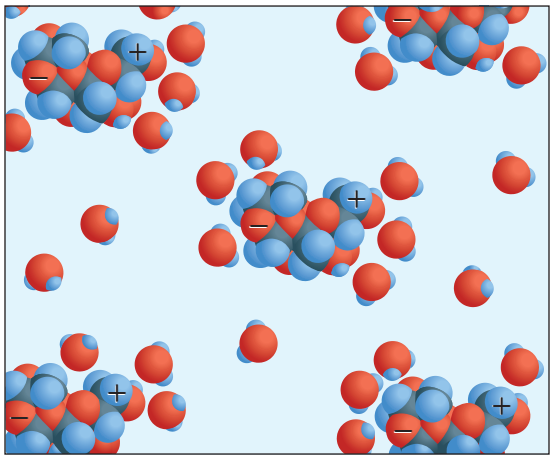
Figure 4 Dissolving sugar in water is a three-step process.



Step 1 At the surface of the sugar crystal, oppositely charged parts of the sugar and water molecules attract each other.



Step 2 Because the water molecules are moving in the liquid, they pull sugar molecules away from the crystal.



Step 3 Water molecules and sugar molecules continue to spread out until a homogeneous mixture forms.

Program: **HSS_NA**	Component: **Physical_Science**	**PDF Pass**
Vendor: **Aptara**	Chapter: **21**	